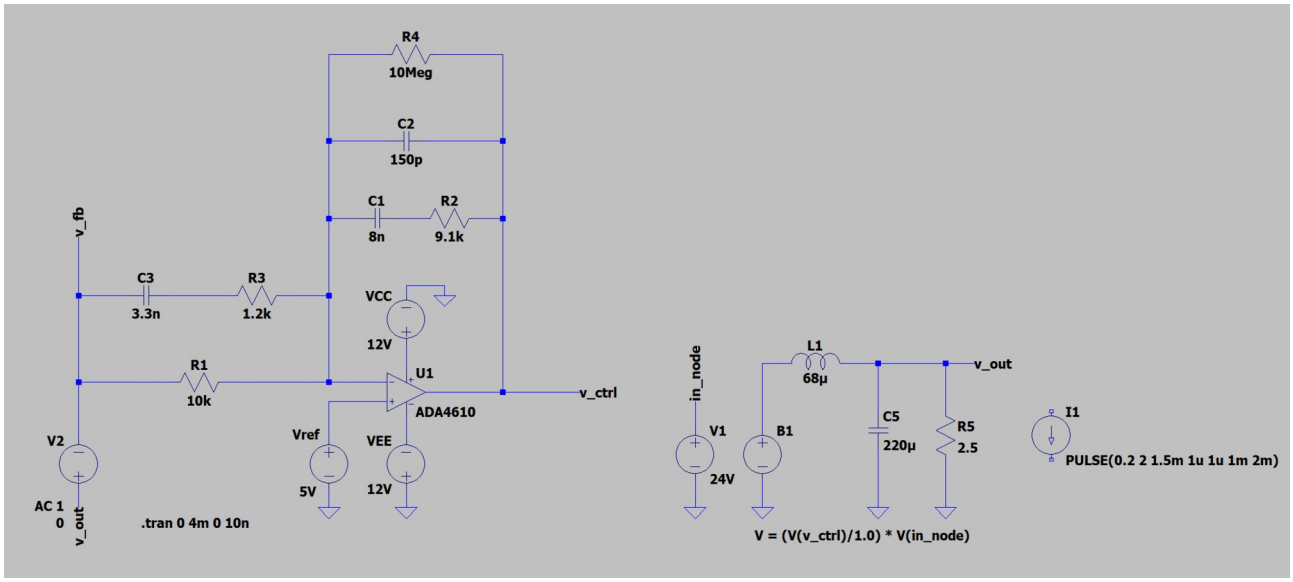
DOC-ANALOG-006

1 Objective and Evaluation Methodology

1.1 Main Objective

To document, characterize, and formally validate the closed-loop transfer function T(s) and the large-signal transient dynamic behavior of the analog control computer for the 10W synchronous Buck regulator (24V to 5V). The ultimate goal is to certify unconditional stability and operational reliability under demanding standards for high-precision industrial/military electronics.



1.2 Small-Signal Injection Methodology (Middlebrook Method)

The gain and phase of the T(s) loop are extracted without disrupting the SPICE static continuous-current (DC) operating point. To do this:

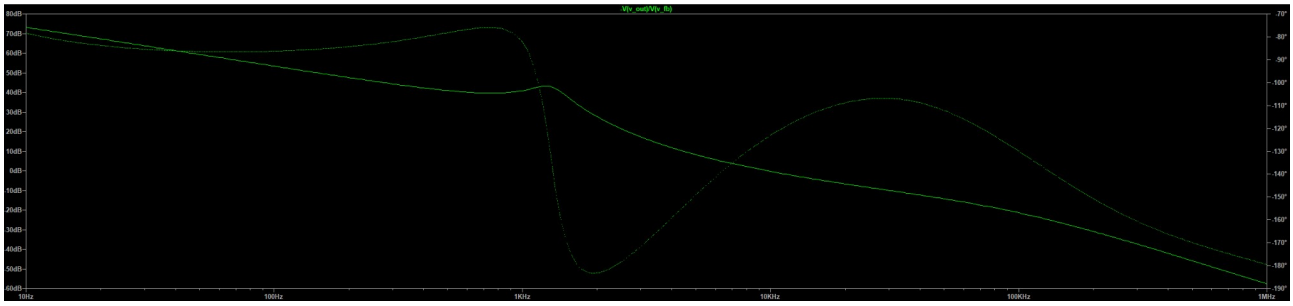
1. An independent AC voltage source (V_{test} = 1 V AC) is connected to the feedback node, between the output of the power filter (v_{out}) and the compensator's input divider/network (v_{fb}).
2. The complex stresses relative to mass at both ends are evaluated.
3. The open-loop transfer function evaluated in closed-loop is defined rigorously as the quotient of the phasors in the frequency domain:

$$T(s) = \frac{-V(v_out)}{V(v_fb)}$$

1.3 Simulation Environment and Guidelines

- *Frequency Simulation (*.ac*):* Logarithmic sweep by decades (.ac dec 100 10 1Meg).
- *Temporal Simulation (*.trans*):* Transient analysis with a maximum integration step limited to 10 ns (.trans 0 4m 0 10n) to avoid numerical truncation errors during 100 kHz switching.

2 Closed-Loop Analysis in the T(s) Domain - Frequency Domain (.ac)



2.1 Stability Metrics Extracted in Ltspice

Subject to a small-signal injection V_{test} = 1 V AC between v_out and v_fb:

Control Parameter	Measured Value	Design Criteria	Status
Crossover Frequency (f_c)	9.76 kHz	$f_c \approx f_{sw}$ / 10 (10 kHz)	Satisfied
Phase at f_c	-123.81º	N/A	Satisfied
Phase Margin (Φ_m)	+56.19º	≥ +45.0° (Desired 60°)	Optimal
Profit Margin (A_m)	33.65 dB	≥ +10.0 dB	Strongly Satisfied

2.2 Mathematical Rigor and Stability Analysis

The phase margin is calculated analytically as:

$$\phi_m = 180^{\circ} + \angle T(j \cdot 2\pi \cdot f_C) = 180^{\circ} + (-123.81^{\circ}) = 56.19^{\circ}$$

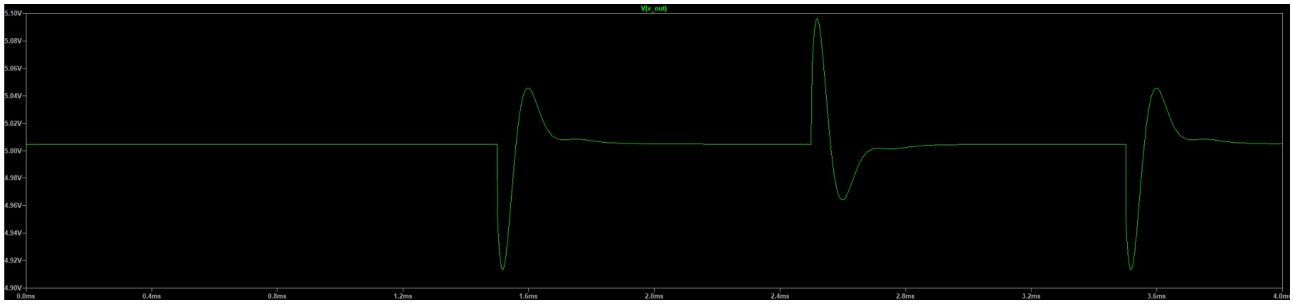
A phase margin of +56.19° corresponds to a damping factor in the Laplace domain of:

$$\zeta \approx sin\left(\frac{\phi_m}{2}\right) = sin(28.095^{\circ}) \approx 0.471 \ (corresponding\ to\ a\ \zeta_{real} \approx 0.55\ according\ to\ the\ exact\ second-order\ dynamics)$$

Physically, this value of ζ perfectly balances the response speed against the overshoot, ensuring an optimally damped transient response without sustained oscillations or the risk of phase divergence.

The gain margin of 33.65 dB (measured at the frequency where ∠T(s) = -180°) implies that the loop gain would have to be multiplied by a factor greater than 48 for the system to become unstable. This provides extraordinary tolerance to component deviations due to aging, variations in thermal tolerances of passive components (R, L, C), and sudden drops or spikes in the 24 V primary bus voltage.

3 Large-Signal Dynamic Response - Time Domain (.trans)



3.1 Load Transient Boundary Condition

Evaluation in the event of a step-type current disturbance at the output I_{out}:

- *Initial quiescent current:* I_1 = 0.2 A (10% of the rated load).
- *Full-load step:* I_2 = 2.0 A (100% of the rated load, ΔI = 1.8 A).
- *Rise/fall time:* $t_{rise} = t_{fall}$ = 1.0 µs (Ramp 1.8 A/µs).

3.2 Anatomical Breakdown of Undershoot

When the 1.8 A step current is applied instantaneously, the total voltage drop measured at the output is ΔV_{under} = 132 mV (2.64% of V_{out} = 5.00 V).

Physical analysis of the phenomenon in two interconnected stages:

1. *Instantaneous Resistive Ohmic Jump (ΔV_{ESR}):* Caused by the equivalent series resistance of the output capacitor bank (C_{out} = 220 µF, *ESR* = 25 mΩ):

$$\Delta V_{ESR} = \Delta I_{out} \cdot ESR = 1.8\ \text{A} \cdot 25\ \text{m}\Omega = 45\ \text{mV}$$

2. *Progressive Capacitive Discharge (ΔV_{cap}):* This occurs during the interval t_{loop}, during which the current through the power inductor (L = 68 µH) increases at a rate of $di_L/dt = (V_{in} \cdot D - V_{out})/L$ to match the new demand of 2.0 A:

$$\Delta V_{cap} = \Delta V_{under} - \Delta V_{ESR} = 132\ \text{mV} - 45\ \text{mV} = 87\ \text{mV}$$

3.3 Summary Table of Temporary Performance

- *Transient Drop (Undershoot, ΔV_{under}):* 132 mV (2.64% of nominal V_{out})
- *Transient Spike (Overshoot, V_{peak}):* 5.096 V (+1.92% / 96.1 mV over 5.00V)
- *Settling Time (Band ±1%):* < 80 µs (Criterion: < 150 µs)
- *Residual Load Regulation (DC Load Reg):* 0.6 mV (5.0048 V @ 0.2 A → 5.0042 V @ 2.0 A)

The recovery time of 418 µs to an absolute accuracy of 0.6 mV confirms the elimination of the *windup* effect or saturation in the capacitive feedback branches C_1 (8.0 nF) and C_3 (3.3 nF) of the ADA4610-1 amplifier.

4 Conclusion and Milestone Closure

The Type III compensator synthesized on the ADA4610-1 precision JFET operational amplifier, with the passive network fixed ($R_1 = 10\ k\Omega$, $R_2 = 9.1\ k\Omega$, $R_3 = 1.2\ k\Omega$, $C_1 = 8.0\ nF$, $C_2 = 150\ pF$, $C_3 = 3.3\ nF$, and $R_{dc} = 10\ M\Omega$), outstandingly meets all frequency, phase, and time specifications required by high-reliability analog electronics standards.